

MAXWELL AFB, AL, USA

WMO: 722265

Lat: 32.383N

Lon: 86.350W

Elev: 52

StdP: 100.70

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.8	(c) -1.8	(d) -12.3	(e) 1.3	(f) 0.4	(g) -9.8	(h) 1.6	(i) 2.1	(j) 9.4	(k) 11.3	(l) 8.6	(m) 10.7	(n) 2.3	(o) 340	(p) 0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	36.2	24.3	35.1	24.6	34.0	24.5	26.7	32.5	26.2	32.0	25.7	31.2	3.0	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	28.7	24.9	20.0	28.4	24.3	19.4	27.9	83.6	32.7	81.4	31.9	79.3	31.3	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	7.0	5.8	-6.9	37.1	2.6	1.5	-8.7	38.1	-10.2	39.0	-11.6	39.8	-13.5	40.9		
			-7.9	27.7	2.6	0.7	-9.8	28.2	-11.3	28.6	-12.8	29.0	-14.7	29.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.2	8.7	11.6	14.9	18.8	23.4	27.1	28.0	25.6	19.8	13.5	10.9			
	DBStd	7.91	5.74	5.21	4.78	3.62	2.94	2.06	1.60	1.87	2.62	4.16	4.36	5.10		
	HDD10.0	217	94	40	14	1	0	0	0	0	0	16	52			
	HDD18.3	1084	300	194	125	38	3	0	0	0	33	155	235			
	CDD10.0	3592	54	86	166	265	417	513	558	469	305	119	81			
	CDD18.3	1418	2	6	19	52	162	263	300	301	219	79	9	5		
	CDH23.3	13861	9	32	154	467	1505	2729	3112	3103	1994	678	54	23		
	CDH26.7	6183	0	2	22	108	602	1339	1476	1511	909	208	6	1		
(14) Wind	WSAvg	2.4	2.8	2.9	3.0	2.7	2.3	2.1	1.9	2.0	2.1	2.3	2.3	2.5		
(15) Precipitation	PrecAvg	1270	115	121	148	111	84	99	128	100	86	63	86	128		
	PrecMax	1766	250	324	331	271	237	246	243	257	237	208	236	347		
	PrecMin	871	38	36	31	39	9	42	58	38	18	0	18	43		
	PrecStd	198	49	58	76	55	55	46	50	43	55	45	50	65		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	26.6	29.0	30.6	34.4	37.2	37.1	38.4	36.3	32.5	27.2	25.9		
		MCWB	18.6	19.9	19.1	20.6	22.6	24.2	25.1	24.1	23.3	22.4	20.1	20.1		
	2%	DB	22.3	24.0	27.0	29.3	32.8	35.7	35.3	36.2	34.9	30.8	24.9	23.6		
		MCWB	17.5	18.7	18.6	20.2	22.4	24.2	25.3	24.3	23.4	21.9	18.5	18.9		
	5%	DB	20.0	22.2	25.0	27.9	31.6	34.4	34.2	34.8	33.4	29.0	23.0	21.5		
		MCWB	16.6	17.1	17.5	19.8	22.1	24.3	25.2	24.9	23.4	20.9	17.6	18.5		
	10%	DB	17.8	20.1	23.0	26.3	30.2	33.0	33.2	33.3	31.8	27.3	21.2	19.3		
		MCWB	15.0	15.9	16.5	18.8	21.6	24.1	25.1	24.9	23.3	20.2	16.5	16.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	21.0	21.2	23.3	25.0	26.9	27.5	27.4	25.8	24.6	22.2	21.8		
		MCDB	22.2	25.0	25.1	28.0	29.9	32.7	33.2	33.2	31.4	28.5	24.1	24.1		
	2%	WB	18.9	19.6	20.2	22.2	24.0	26.1	26.8	26.7	25.2	23.8	20.6	20.3		
		MCDB	21.2	22.8	24.2	26.6	29.7	32.1	32.5	32.6	30.6	27.8	23.2	22.2		
	5%	WB	17.4	18.6	19.1	21.2	23.3	25.6	26.2	26.2	24.8	23.0	19.1	18.9		
		MCDB	19.1	21.5	23.3	25.3	28.8	31.5	31.8	32.0	29.8	26.9	21.7	20.9		
	10%	WB	15.8	17.2	18.0	20.3	22.8	25.1	25.7	25.7	24.3	22.1	17.6	17.1		
		MCDB	17.8	19.2	21.8	24.3	28.0	30.8	31.0	30.9	29.0	25.5	20.0	19.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.7	12.2	12.4	12.0	11.1	10.2	10.5	11.1	12.2	12.3	10.6		
		MCDBR	13.1	12.8	13.9	13.8	13.6	12.8	11.6	12.3	13.2	13.0	13.2	12.3		
	5% WB	MCWBR	8.5	7.8	6.8	6.0	4.7	3.8	3.6	3.5	3.8	5.1	7.6	8.4		
		MCWBR	10.2	10.6	11.2	10.9	10.9	11.0	10.4	10.5	10.3	10.2	10.7	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.324	0.332	0.361	0.392	0.442	0.481	0.538	0.528	0.434	0.374	0.347	0.327		
	taud		2.518	2.501	2.425	2.331	2.236	2.179	2.027	2.051	2.299	2.445	2.462	2.523		
	Ebn at Noon		898	926	918	899	851	815	767	769	834	866	861	874		
	Edh at Noon		85	96	111	127	141	149	172	166	125	100	90	81		
(44) All-Sky Solar Radiation	RadAvg		2.79	3.36	4.52	5.65	6.22	6.17	5.99	5.55	4.92	4.18	3.15	2.39		
	RadStd		0.19	0.41	0.41	0.40	0.44	0.46	0.41	0.32	0.42	0.52	0.30	0.26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+161	+105

Nomenclature: See separate page